Integration by Parts

ANSWER KEY

The integration by parts formula

1 State the integration by parts formula, and explain the LIATE guideline for choosing u.
$\int u\,dv = uv - \int v\,du$. LIATE suggests choosing u in the priority order: Logarithmic, Inverse trig, Algebraic, Trigonometric, Exponential — the function type listed first should typically be u.

2 Derive the integration by parts formula starting from the product rule $\frac{d}{dx}[uv] = u'v + uv'$.
Integrating both sides: $uv = \int u'v\,dx + \int uv'\,dx = \int v\,du + \int u\,dv$. Rearranging: $\int u\,dv = uv - \int v\,du$.

3 Evaluate $\int xe^x\,dx$.
Let $u = x$, $dv = e^x\,dx$: $du = dx$, $v = e^x$. $\int xe^x\,dx = xe^x - \int e^x\,dx = xe^x - e^x + C$.

4 Evaluate $\int x\cos x\,dx$.
Let $u = x$, $dv = \cos x\,dx$: $du = dx$, $v = \sin x$. $\int x\cos x\,dx = x\sin x - \int \sin x\,dx = x\sin x + \cos x + C$.

Integration by parts with logarithms and inverse trig

5 Evaluate $\int \ln x\,dx$.
Let $u = \ln x$, $dv = dx$: $du = \frac{1}{x}dx$, $v = x$. $\int \ln x\,dx = x\ln x - \int 1\,dx = x\ln x - x + C$.

6 Evaluate $\int x\ln x\,dx$.
Let $u = \ln x$, $dv = x\,dx$: $du = \frac{1}{x}dx$, $v = \frac{x^2}{2}$. $\int x\ln x\,dx = \frac{x^2}{2}\ln x - \int \frac{x}{2}\,dx = \frac{x^2}{2}\ln x - \frac{x^2}{4} + C$.

7 Evaluate $\int \tan^{-1} x\,dx$.
Let $u = \tan^{-1} x$, $dv = dx$: $du = \frac{1}{1+x^2}dx$, $v = x$.
$\int \tan^{-1} x\,dx = x\tan^{-1} x - \int \frac{x}{1+x^2}\,dx = x\tan^{-1} x - \frac{1}{2}\ln(1+x^2) + C$.

8 Evaluate $\int x^2\ln x\,dx$.
Let $u = \ln x$, $dv = x^2\,dx$: $du = \frac{1}{x}dx$, $v = \frac{x^3}{3}$. $\int x^2\ln x\,dx = \frac{x^3}{3}\ln x - \int \frac{x^2}{3}\,dx = \frac{x^3}{3}\ln x - \frac{x^3}{9} + C$.

9 Evaluate $\int \sin^{-1} x\,dx$.
Let $u = \sin^{-1} x$, $dv = dx$: $du = \frac{1}{\sqrt{1-x^2}}dx$, $v = x$.
$\int \sin^{-1} x\,dx = x\sin^{-1} x - \int \frac{x}{\sqrt{1-x^2}}\,dx = x\sin^{-1} x + \sqrt{1-x^2} + C$.

Repeated integration by parts

10 Evaluate $\int x^2e^x\,dx$, applying integration by parts twice.
First: $u = x^2$, $dv = e^x\,dx$: $\int x^2e^x\,dx = x^2e^x - 2\int xe^x\,dx$. Using the earlier result $\int xe^x\,dx = xe^x - e^x$:
$= x^2e^x - 2(xe^x - e^x) + C = x^2e^x - 2xe^x + 2e^x + C$.

11 Evaluate $\int x^2 \sin x\,dx$, applying integration by parts twice.

First: $u = x^2$, $dv = \sin x\,dx$: $= -x^2\cos x + 2\int x\cos x\,dx$. Using the earlier result $\int x\cos x\,dx = x\sin x + \cos x$: $= -x^2\cos x + 2x\sin x + 2\cos x + C$.

12 Set up (do not evaluate fully) the reduction needed to find $\int x^3 e^x\,dx$, identifying how many times integration by parts must be applied.

Each application of IBP with $u = x^n$ reduces the power of x by 1 while v stays an antiderivative of e^x. Since the power starts at 3, integration by parts must be applied 3 times to eliminate the polynomial factor entirely.

The cyclic case

13 Evaluate $\int e^x \sin x\,dx$ using the cyclic (solve-for-the-integral) technique.

Let $I = \int e^x \sin x\,dx$. First IBP ($u = \sin x$, $dv = e^x dx$): $I = e^x \sin x - \int e^x \cos x\,dx$. Second IBP on the remaining integral ($u = \cos x$, $dv = e^x dx$): $\int e^x \cos x\,dx = e^x \cos x + \int e^x \sin x\,dx = e^x \cos x + I$. So $I = e^x \sin x - (e^x \cos x + I) \Rightarrow 2I = e^x \sin x - e^x \cos x \Rightarrow I = \frac{e^x(\sin x - \cos x)}{2} + C$.

14 Evaluate $\int e^{2x} \cos x\,dx$ using the cyclic technique.

Let $I = \int e^{2x} \cos x\,dx$. IBP ($u = \cos x$, $dv = e^{2x} dx$): $I = \frac{1}{2}e^{2x}\cos x + \frac{1}{2}\int e^{2x}\sin x\,dx$. IBP again on that integral ($u = \sin x$, $dv = e^{2x}dx$): $\int e^{2x}\sin x\,dx = \frac{1}{2}e^{2x}\sin x - \frac{1}{2}I$. Substituting: $I = \frac{1}{2}e^{2x}\cos x + \frac{1}{4}e^{2x}\sin x - \frac{1}{4}I \Rightarrow \frac{5}{4}I = \frac{1}{2}e^{2x}\cos x + \frac{1}{4}e^{2x}\sin x \Rightarrow I = \frac{e^{2x}(2\cos x + \sin x)}{5} + C$.

Definite integrals using integration by parts

15 Evaluate $\int_0^1 xe^x\,dx$ using the antiderivative found earlier.

$[xe^x - e^x]_0^1 = (e - e) - (0 - 1) = 0 - (-1) = 1$.

16 Evaluate $\int_1^e \ln x\,dx$ using the antiderivative found earlier.

$[x\ln x - x]_1^e = (e \cdot 1 - e) - (0 - 1) = 0 - (-1) = 1$.

17 Evaluate $\int_0^\pi x\sin x\,dx$.

Let $u = x$, $dv = \sin x\,dx$: $\int x\sin x\,dx = -x\cos x + \int \cos x\,dx = -x\cos x + \sin x + C$.

$[-x\cos x + \sin x]_0^\pi = (-\pi(-1) + 0) - (0 + 0) = \pi$.

Integration by parts to derive a reduction formula

18 Use integration by parts to show $\int x^n e^x\,dx = x^n e^x - n\int x^{n-1} e^x\,dx$ (the reduction formula), and use it once to reconfirm $\int x^2 e^x\,dx$.

Let $u = x^n$, $dv = e^x\,dx$: $du = nx^{n-1}dx$, $v = e^x$. $\int x^n e^x\,dx = x^n e^x - n\int x^{n-1}e^x\,dx$, confirming the reduction formula. With $n = 2$: $\int x^2 e^x\,dx = x^2 e^x - 2\int xe^x\,dx = x^2e^x - 2xe^x + 2e^x + C$, matching the earlier result.

Choosing u and dv wisely

19 For each integral, state the best choice of u (using LIATE) without evaluating:

a $\int x^3 \ln x\,dx$ $u = \ln x$ (Logarithmic beats Algebraic)

b $\int xe^{5x}\,dx$ $u = x$ (Algebraic beats Exponential)

c $\int x\tan^{-1} x\,dx$ $u = \tan^{-1} x$ (Inverse trig beats Algebraic)

d $\int x^2 \cos x\,dx$ $u = x^2$ (Algebraic beats Trigonometric)

20 Explain what goes wrong if you attempt $\int xe^x\,dx$ with the reversed choice $u = e^x$, $dv = x\,dx$.

Then $du = e^x\,dx$, $v = \frac{x^2}{2}$, giving $\int xe^x\,dx = \frac{x^2}{2}e^x - \int \frac{x^2}{2}e^x\,dx$. The remaining integral is MORE complicated than the original (higher power of x under the same exponential), so this choice makes the problem worse rather than simpler — LIATE's algebraic-before-exponential guidance avoids this trap.

Integration by parts with algebraic-trigonometric products

21 Evaluate $\int x\sec^2 x\,dx$.

Let $u = x$, $dv = \sec^2 x\,dx$: $du = dx$, $v = \tan x$.
$\int x\sec^2 x\,dx = x\tan x - \int \tan x\,dx = x\tan x + \ln|\cos x| + C$.

22 Evaluate $\int 2x\sin(3x)\,dx$.

Let $u = 2x$, $dv = \sin(3x)\,dx$: $du = 2\,dx$, $v = -\frac{1}{3}\cos(3x)$.
$\int 2x\sin(3x)\,dx = -\frac{2x}{3}\cos(3x) + \frac{2}{3}\int \cos(3x)\,dx = -\frac{2x}{3}\cos(3x) + \frac{2}{9}\sin(3x) + C$.

Definite integrals: area under a transcendental product

23 Find the exact area under $y = xe^{-x}$ on $[0, 3]$, using $\int xe^{-x}\,dx = -xe^{-x} - e^{-x} + C$ (verify this antiderivative first by differentiating).

Check: $\frac{d}{dx}[-xe^{-x} - e^{-x}] = -e^{-x} + xe^{-x} + e^{-x} = xe^{-x}$ ✓. Area
$= [-xe^{-x} - e^{-x}]_0^3 = (-3e^{-3} - e^{-3}) - (0 - 1) = 1 - 4e^{-3} \approx 1 - 0.199 \approx 0.801$.

24 Evaluate $\int_0^{\pi/2} x\cos x\,dx$ using the antiderivative $x\sin x + \cos x$ found earlier.

$[x\sin x + \cos x]_0^{\pi/2} = \left(\frac{\pi}{2}(1) + 0\right) - (0 + 1) = \frac{\pi}{2} - 1 \approx 0.571$.

Mixed review: choosing the right technique

25 For each integral, state whether it should be solved by u-substitution or integration by parts, and identify the key choice (substitution variable or u) without fully evaluating:

a $\int 2xe^{x^2}\,dx$ Substitution ($w = x^2$) — the derivative of the exponent is present as a factor

b $\int x^2 e^x\,dx$ Integration by parts ($u = x^2$, twice) — no matching derivative factor for substitution

c $\int \frac{\ln x}{x}\,dx$ Substitution ($w = \ln x$) — $\frac{1}{x}$ is exactly dw

d $\int x \ln x\,dx$ Integration by parts ($u = \ln x$) — no simplifying substitution exists here